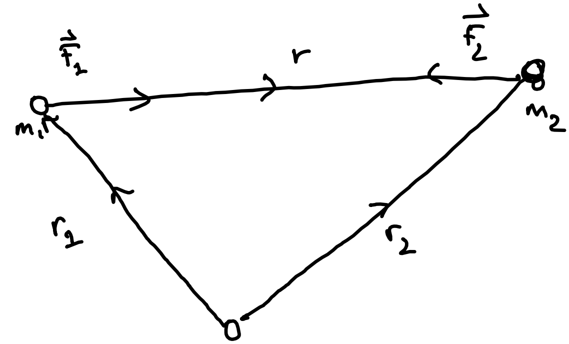
Two body problem:



$$\vec{F}_{1,2} = \frac{G m_1 m_2 \vec{r}}{r^3} = m_{1,2}\ddot{\vec{r}}_{1,2}$$

$$m_1\ddot{\vec{r}}_1 + m_2\ddot{\vec{r}}_2 = 0$$ ⟶ can be integrated twice to give the equations of motion

Simplify considerably if we consider the motion of m_2 relative to m_1, then

$$\ddot{\vec{r}} + \mu\frac{\vec{r}}{r^3} = 0 \,;\ \mu = G(m_1+m_2)$$

⟶ this describes the motion of the Earth around the Sun.

⟶ if you follow the process, accounting for eccentricity, you can find the equation of motion, or Kepler's equation

$$r = \frac{a(1-e^2)}{1+e\cos(f\ )},$$

a: semimajor axis

θ: true longitude, specifies the direction to the observer

ϖ: longitude of pericenter

f: θ-ϖ: true anomaly

* Elliptic equations are usually solved numerically

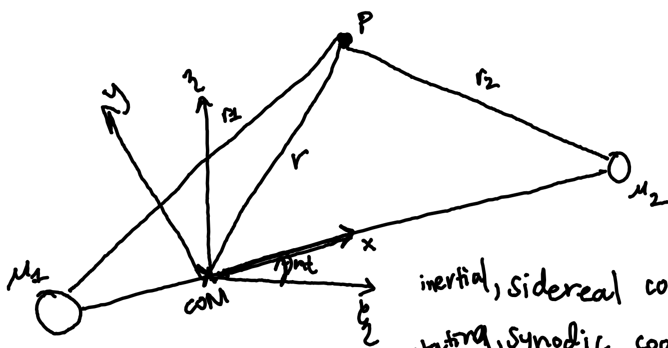
The two body problem produces closed orbits, what about adding a 3rd body?

$\longrightarrow$ people have been working on the 3 body problem for centuries; there's also a novel guess

Restricted 3 body Problem:

assume: two bodies orbit in circular, coplanar orbits and that the mass of the third body is too small to affect the motion of the more massive bodies.

$\longrightarrow$ works well for Sun - planet - satellite



inertial, sidereal coordinates: $\overbrace{(\xi, \eta, \zeta)}^{\text{non rotating}}$

corotating, synodic coordinates : (x, y, z)

ζ & z are axes of rotation out of the page.

let $\mu = G(m_1 + m_2) = 1$

if $m_1 > m_2$, $\bar{\mu} = m_2/(m_1+m_2)$

$\longrightarrow \mu_1 = Gm_1 = 1 - \bar{\mu}$, $\mu_2 = Gm_2 = \bar{\mu}$, $\bar{\mu} < 1/2$

$$\ddot{\xi} = \mu_1 \frac{\xi_1 - \xi}{r_1^3} + \mu_2 \frac{\xi_2 - \xi}{r_2^3} , \quad \ddot{\eta} = \mu_1 \frac{\eta_1 - \eta}{r_1^3} + \mu_2 \frac{\eta_2 - \eta}{r_2^3}$$

$$\ddot{\zeta} = \mu_1 \frac{\zeta_1 - \zeta}{r_1^3} + \mu_2 \frac{\zeta_2 - \zeta}{r_2^3}$$

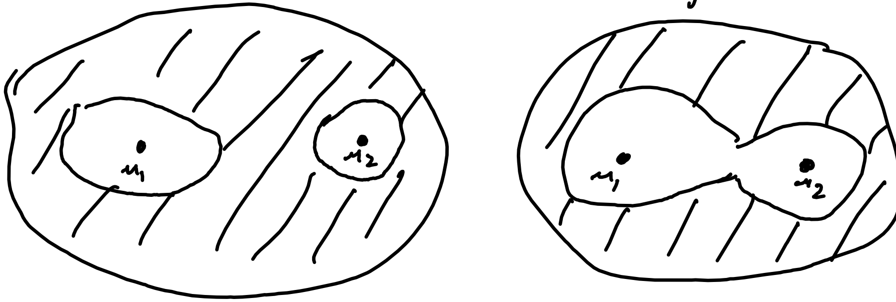
if x-axis is chosen such that

$$(x_1, y_1, z_1) = (-\mu_2, 0, 0) \; ; \; (x_2, y_2, z_2) = (\mu_1, 0, 0)$$

$$\begin{pmatrix} \xi \\ \eta \\ \zeta \end{pmatrix} = \begin{pmatrix} \cos nt & -\sin nt & 0 \\ \sin nt & \cos nt & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

can be written as gradient of scalar function

⟶ solved w/ Jacobi integral w/ integration constant C_J ↓ that you need to solve for



⟶ based on integration constant and masses.

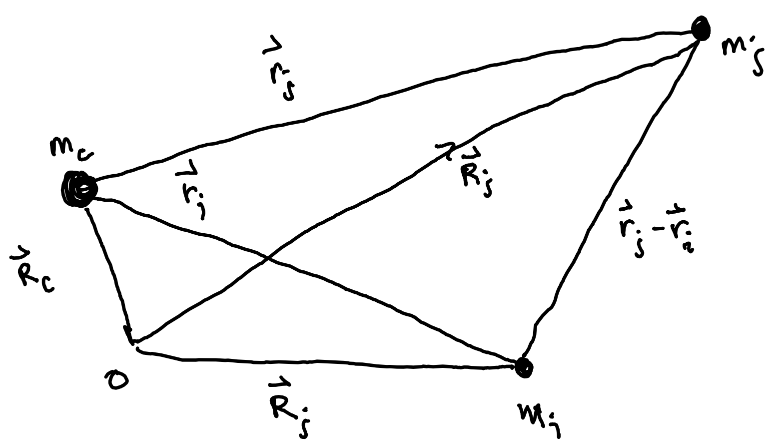
⟶ can also use to determine Lagrange equilibrium points and motion around them

⟶ Trojan asteroids orbit L4, L5

$M_\odot \cong 1000\, M_J \cong 10^{11} M_{\text{Trojans}}$

when we have 3 bodies of appreciable mass, the problem is not integrable and must be solved numerically.

In the case where the motion is dominated by a central body, and the other two orbits are perturbations to this motion we can make some headway. The additional accelerations felt by the two perturbing bodies is described by the DISTURBING FUNCTION $\longrightarrow$ key to planetary dynamics



little $\vec{r}$'s are wrt m_c while big $\vec{R}$'s are wrt to an origin in the inertial reference frame

$|\vec{r}_i| = r_i = (x_i^2 + y_i^2 + z_i^2)^{1/2}$; same for $|\vec{r}_j|$

$$|\vec{r}_j - \vec{r}_i| = \left[(x_j - x_i)^2 + (y_j - y_i)^2 + (z_i - z_j)^2\right]^{1/2}$$

can write down the EOM in the inertial reference frame.

$$M_c \ddot{\vec{R}}_c = G m_c m_i \frac{\vec{r}_i}{r_i^3} + G m_c m_j \frac{\vec{r}_j}{r_j^3}$$

$$m_i \ddot{\vec{R}}_i = G m_i m_j \frac{(\vec{r}_j - \vec{r}_i)}{|\vec{r}_j - \vec{r}_i|^3} - \frac{G M_i m_c \vec{r}_i}{r_i^3}$$

$$m_j \ddot{\vec{R}}_j = G m_i m_j \frac{(\vec{r}_i - \vec{r}_j)}{|\vec{r}_i - \vec{r}_j|^3} - G m_j m_c \frac{\vec{r}_j}{r_j^3}$$

in this case:

$$\ddot{\vec{r}}_i = \ddot{\vec{R}}_i - \ddot{\vec{R}}_c \; ; \; \ddot{\vec{r}}_j = \ddot{\vec{R}}_j - \ddot{\vec{R}}_c$$

plug in $\ddot{\vec{R}}_i$, $\ddot{\vec{R}}_c$, $\ddot{\vec{R}}_j$ to get:

$$\ddot{\vec{r}}_i + G(m_c + m_i)\frac{\vec{r}_i}{r_i^3} = G m_j \left(\frac{\vec{r}_j - \vec{r}_i}{|\vec{r}_j - \vec{r}_i|^3} - \frac{\vec{r}_j}{r_j^3}\right)$$

$$\ddot{\vec{r}}_j + G(m_c + m_j)\frac{\vec{r}_j}{r_j^3} = G m_i \left(\frac{\vec{r}_i - \vec{r}_j}{|\vec{r}_j - \vec{r}_i|^3} - \frac{\vec{r}_i}{r_i^3}\right)$$

These equations can be written in terms of gradients
↳ for relative acceleration

of scalar functions:

$$\ddot{\vec{r}}_i = \nabla_i (\mathcal{U}_i + R_i) = \left(\hat{i}\frac{\partial}{\partial x_i} + \hat{j}\frac{\partial}{\partial y_i} + \hat{k}\frac{\partial}{\partial z_i}\right)(\mathcal{U}_i + R_i)$$

&

$$\ddot{\vec{r}}_j = \nabla_j (\mathcal{U}_j + R_j) = \left(\hat{i}\frac{\partial}{\partial x_j} + \hat{j}\frac{\partial}{\partial y_j} + \hat{k}\frac{\partial}{\partial z_j}\right)(\mathcal{U}_j + R_j)$$

here:

$$\mathcal{U}_i = G\frac{m_c + m_i}{r_i} \quad \& \quad \mathcal{U}_j = G\frac{m_c + m_j}{r_j}$$

are the central, two body, parts of the potential

and R_i, R_j are the disturbing functions.

Since $\ddot{\vec{r}}_i$ & $\ddot{\vec{r}}_j$ are not functions of each other's x_i, y_i, z_i

$$R_i = \frac{Gm_j}{|\vec{r}_j - \vec{r}_i|} - Gm_j\frac{\vec{r}_i \cdot \vec{r}_j}{r_j^3}$$

$$R_j = \underbrace{\frac{Gm_i}{|\vec{r}_i - \vec{r}_j|}}_{\text{direct terms}} - \underbrace{Gm_i\frac{\vec{r}_i \cdot \vec{r}_j}{r_i^3}}_{\text{indirect terms}}$$

↳ disappears if origin is at center of mass

Note: tides, spins could add to the disturbing functions.

People often rewrite in planetary terms with "inner" and "outer" secondary.

m, r : inner

m', r' : outer $\quad r' > r$

$$\Rightarrow \ddot{\vec{r}} + G(m_c + m)\frac{\vec{r}}{r^3} = Gm'\left(\frac{\vec{r}' - \vec{r}}{|\vec{r}' - \vec{r}|^3} - \frac{\vec{r}'}{r'^3}\right)$$

$$R = \frac{\mu'}{|\vec{r}' - \vec{r}|} - \mu' \frac{\vec{r} \cdot \vec{r}'}{r'^3}$$

$\Rightarrow \mu' = Gm'$; osculating element:
$n^2 a^3 = G(m_c + m')$

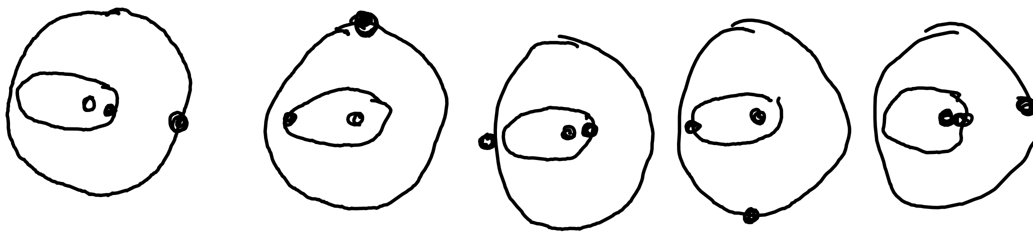
In this case, there are two ways to think about the orbit. There is the motion due to the disturbing function and the "osculating" orbit which describes the motion if the perturbation were to disappear.

This problem is solved by expansion of Legendre Polynomials. $\rightarrow$ chapter 6

The n's are resonant integers and they persist in nature.

These resonances define what remains stable on "secular" or long-term timescales.

Example: 2:1 orbital period ratio



avoids closest approach

if the conjunctio happens at closest approach, perturbations can drive system out of stability.

if each satellite has mean motion, n & n' then

$$\frac{n'}{n} = \frac{p}{p+q}$$ where p & q are integers.

→ if p is 1 and q is 1 : 2:1 resonance
p is 1 and q is 2 : 3:1 resonance